RIZQI ASY'ARI RAMADHAN

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SUMMARY

Agricultural Engineering graduate from Universitas Gadjah Mada with experience in crop production, agricultural operations, technical monitoring, and data analysis. Experienced in IoT-based irrigation systems, sensor testing and calibration, data analysis, and environmental monitoring across plant factory and smart greenhouse environments. Interested in applying agricultural technology, analytical problem-solving, and process improvement to support efficient and sustainable operations.

WORKING EXPERIENCE

Smart Agriculture UGM, Yogyakarta — Oct 2026 – Present

Research Assistant

- Monitored the field implementation of an AURORA fertigation system development project at KWT Teratai Mekar, including project progress, system installation, and field conditions to ensure implementation remained aligned with project requirements.
- Conducted field observations, documented project progress, and identified technical issues during implementation, providing findings and feedback to support project evaluation, troubleshooting, and subsequent system improvements.

Plant Factory Faculty of Agricultural Technology, Yogyakarta — Jul 2026 – Present

Plant Factory Operations Coordinator

- Coordinated daily operations for a 7-member team in a plant factory with a capacity of 480 plants, managing work schedules, task delegation, and facility operations to meet operational standards.
- Monitored production activities and operational conditions, identified issues affecting workflow, and coordinated follow-up actions with team members to maintain consistent plant factory operations and improve work efficiency.

Plant Factory Faculty of Agricultural Technology, Yogyakarta — Aug 2025 – Jun 2026

Plant Factory Grower

- Managed the cultivation of up to 480 plants throughout the production cycle, from seedling preparation and transplanting to plant maintenance and harvesting, while monitoring plant growth and health.
- Monitored and maintained plant factory environmental conditions, including temperature, humidity, lighting and other growing conditions, to maintain an optimal controlled environment for plant growth and production.

PT Mitra Sejahtera Membangun Bangsa, Yogyakarta — Dec 2024 – Feb 2025

Internship – Plantation Smart Greenhouse

- Supported plantation operations in a 400 m² smart greenhouse with a population of approximately 1,200 premium melon crops.
- Controlled greenhouse environmental parameters, including temperature, RH, pH, and EC, maintaining growing conditions within target ranges to support consistent melon development and production.

EDUCATION

Universitas Gadjah Mada, Yogyakarta — Aug 2022 – Jul 2026

Bachelor of Agricultural Technology, Agricultural Engineering

GPA 3.21 of 4.00

Thesis Title: Development of a Hysteresis Based Automatic Sprinkler Irrigation System for Water Use Efficiency on Coastal Sandy Land

PROJECT

Development of IoT-Based Smart Fertigation System, Yogyakarta Sep 2026 – Present

Field Research Assistant

- Surveyed KWT Teratai Mekar and conducted interviews with members to identify cultivation challenges and fertigation needs.
- Evaluated field conditions and monitored smart fertigation system performance to support improved crop productivity.

Development of Automated Sprinkler Irrigation System, Yogyakarta Sep 2025 – Dec 2025

Researcher

- Implemented and evaluated an IoT-based automatic sprinkler irrigation system using three soil moisture sensors and hysteresis control for coastal sandy land.
- Analyzed sensor calibration and irrigation performance, achieving an average R^2 of 0.9437 and MAPE of 1.33%, with 16.91% potential water savings compared with conventional irrigation.

Liquid Fertilizer Mixing Machine Design, Yogyakarta Mar 2025 – Jun 2025

IoT & System Integration Engineer

- Developed the proposed IoT-based monitoring mechanism using a turbidity sensor to assess fertilizer mixing homogeneity and support automated mixing termination.
- Design and fabrication of a 240 L-capacity liquid fertilizer mixing machine, incorporating a 125 W motor, stainless steel propeller impeller, 36 RPM target speed, and calculated shaft torque of 24.5 N·m.

ORGANIZATIONAL & VOLUNTEER EXPERIENCE

Agrotropikal National Seminar, East Kalimantan Apr 2026

Presenter

- Presented research findings on an IoT-based automatic sprinkler irrigation system, explaining its design, implementation, and performance to an academic audience at a national seminar.
- Disseminated research findings demonstrating reliable soil-moisture sensing, with average R^2 0.9437, RMSE 0.40, and MAPE 1.33%.

PERTETA National Seminar, Yogyakarta Sep 2024

Logistics Division

- Coordinated logistics operations for the PERTETA National Seminar, ensuring smooth event execution for 300+ participants.
- Managed event logistics and resource distribution, supporting timely setup and efficient coordination across event activities.

PERMATETA UGM, Yogyakarta Feb 2023 – Feb 2025

Staff of communication and information

- Managed Instagram account with 2,000+ followers, maintaining consistent audience engagement.
- Planned and published ~10 contents/month through content brainstorming and scheduling.

LANGUAGES AND SKILLS

- **Languages**: Bahasa Indonesia (Native), English (Intermediate)
- **Technical Skills**: Agricultural Production & Crop Management, Agronomy, IoT & Automation, Data Collection & Analysis, Greenhouse Environmental Monitoring, Research & Experimental Testing.
- **Soft Skills**: Analytical Thinking, Problem Solving, Team Coordination, Operational Monitoring, Adaptability
- **Interest**: Smart & Precision Agriculture, Sustainable Production Systems, Greenhouse Technology, Agronomy, Crop Production, Supply Chain.